feb. 3

- Last time: Socks & Shoes Thm $(ab)^{-1} = b^{-1}a^{-1}$
  - Proof Compute $(b^{-1}a^{-1}) \cdot ab = b^{-1}a^{-1}ab = b^{-1}eb = b^{-1}b = e$.

    We only proved left inverse, not necessarily right inverse

    Right inverse: If $cd=e$, then $c=d^{-1}$

    Proof: Suppose $c,d \in G$ with $cd=e$. Group operation by d^{-1} on the right, $cdd^{-1} = ed^{-1}$. Thus $c = d^{-1}$. $\square$

    $(cd)d^{-1} = (e)d^{-1}$. By assoc. & identity, $c(dd^{-1}) = d^{-1}$. By inverses, $c(e) = d^{-1}$, and by identity, $c = d^{-1}$.

  following the socks & shoes proof, other half computes $ab(b^{-1}a^{-1}) = abb^{-1}a^{-1} = aea^{-1} = aa^{-1} = e$

  Note: group axioms state if $d \in G$, then $\exists\, d^{-1} \in G$ s.t. $dd^{-1} = d^{-1}d = e$.

  Does this mean $cd=e \Rightarrow c=d^{-1}$? the inverse exists

- In homework proofs, generally better to have too much detail than not enough
  - Can write in column setup, but justify steps in words
  - Should be able to read aloud in coherent, complete sentences

- Canvas announcement
  - Changed office hrs & HW due date for this week & next week

- D_3 = The group of symmetries of a $\triangle$
  - $\langle r,s : r^3 = s^2 = e \ \& \ rs = sr^2 \rangle$ ← generators & relations

    this tells you everything you need to know to derive all relations
- D_n = The group of symmetries of a regular n-gon
  - Also called the dihedral group on n vertices
  - Presentation in generators & relations $= \langle r,s : r^n = s^2 = e \ \& \ rs = sr^{n-1} \rangle$

    rotation by $\frac{360°}{n}$ flip across vertical axis
- Dihedral groups exist in higher dimensions, not covered in this course

Ch. 3: Finite Groups & Subgroups

- The number of elements in a group G is called its order
  - We denote $|G|$

Ex $|\mathbb{Z}|=\infty$, $|\mathbb{Z}/2\mathbb{Z}|=2$, $|D_3|=6$, $|D_4|=8$, $|\mathbb{Z}/17\mathbb{Z}|=17$

- If a is an element of a group G, we define the order of a to be the least positive integer n such that $\underbrace{a\cdot a\cdot\ldots\cdot a}_{n \text{ times}}=a^n=e$. We denote this $|a|=n$. If no such integer exists, we say the order of a is infinite and write $|a|=\infty$.
  - for $n\in\mathbb{N}^+$, define $a^n=\underbrace{a\cdot a\cdot\ldots\cdot a}_{n \text{ times}}$. Define $a^0=e$ and $a^{-n}=\underbrace{a^{-1}\cdot\ldots\cdot a^{-1}}_{n \text{ times}}$
  - fact: exponent rules work for groups. $a^n\cdot a^m=a^{n+m}$, $(a^n)^m=a^{nm}$

Ex Let $r\in D_3$, $|r|=3$. What about $|rs|$?
$(rs)^2=rsrs=sr^2rs=sr^3s=ss=e$, so $|rs|=2$

- Note: when we refer to $\mathbb{Z}$ as a group, we generally mean under $+$
  - Under mult., $\mathbb{Z}^\times=\{+1,-1\}$ = the multiplicative group of the integers

- Consider
  - $1\in(\mathbb{Z},+)\Rightarrow|1|=\infty$ (since $1^n\neq 0$)
  - $\frac{1}{3}\in(\mathbb{Q},+)\Rightarrow|\frac{1}{3}|=\infty$ (since $\underbrace{\frac{1}{3}\cdot\frac{1}{3}\cdot\ldots\cdot\frac{1}{3}}_{n \text{ times}}\neq 0$ for any positive n)

- What is $|\bar{2}|$ where $\bar{2}\in\mathbb{Z}/6\mathbb{Z}$?
  - $\bar{2}\neq\bar{0}$, $\bar{2}+\bar{2}=\bar{4}\neq\bar{0}$, $\bar{2}+\bar{2}+\bar{2}=\bar{6}=\bar{0}$, Thus $|\bar{2}|=3$

- Conjecture: if $\gcd(a,m)=1$, then $|\bar{a}|$ in $\mathbb{Z}/m\mathbb{Z}$ is m. i.e $|\bar{a}_m|=m$

- We can see that if e is the identity of a group G, then $|e|=1$

- Def A subgroup H of a group G is a subset of G that is a group under the operation of G. We denote this by $H\leq G$

- Why do we have order of an element and order of a group?
  - The order of an element $a\in G$ is the order of the smallest subgroup containing a

Ex Consider $H=\{r,r^2,e\}\subset D_3$
Claim: $H<D_3$

- Associative: yes, inherited from D_3
- Identity: yes, $e \in H$
-

	e	r	r^2
e	e	r	r^2
r	r	r^2	e
r^2	r^2	e	r

Shows closure (all results are in H)

Inverses ✓

Thus, we see $H < D_3$

★ Since $|r| = 3 = |H|$, H is also the smallest subgroup containing r

★ additionally, symmetry across the diagonal shows commutativity, so H is also Abelian

Ex. Consider $6 \in (\mathbb{Z}, +)$. What is the smallest subgroup of $\mathbb{Z}$ containing 6?

Let's call this the <u>smallest subgroup generated by 6</u>, denoted $\langle 6 \rangle$

$\langle 6 \rangle = \{\ldots -18, -12, -6, 0, 6, 12, 18, 24 \ldots\} = 6\mathbb{Z}$ = the multiples of 6

we compute

$|\langle 6 \rangle| = \infty = |6|$